

PlantSeg3D: Gaussian Plant Instance Segmentation with Geometry-Aware Refinement

Hinano Shimbo and Meng-Yu Jennifer Kuo

Nara Women’s University, Nara, Japan

Abstract. This paper proposes an annotation-free framework for 3D plant instance segmentation. Existing methods either struggle with overlapping leaves or require costly 3D annotations. Our approach lifts pre-trained SAM 3 masks into a 3D Gaussian Splatting representation through occlusion-aware multi-view voting and refines them with semantic and geometric priors to recover coherent leaf instances. Experiments demonstrate robust segmentation of touching, folded, and occluded leaves without 3D supervision, providing a practical foundation for semantic 3D plant phenotyping.

Keywords: 3D Instance Segmentation · Plant Phenotyping

1 Introduction

Segmenting plants into individual leaves in 3D is fundamental to plant phenotyping, enabling the measurement of leaf count, area, and inclination for growth monitoring, yield estimation, and disease diagnosis [2, 3]. Although 3D Gaussian Splatting (3DGS) [5] reconstructs plant geometry with remarkable fidelity, it lacks leaf-level semantics. Recovering leaf instances remains challenging due to overlap and self-occlusion. Density-based clustering [9, 12] struggles to separate touching leaves, while learning-based methods [4, 6, 11] require labor-intensive 3D annotations [8] and generalize poorly across species and growth stages.

We propose an annotation-free framework for 3D plant instance segmentation that requires neither 3D annotations nor task-specific training. Pretrained 2D instance masks from SAM 3 [1] are lifted onto the 3D Gaussian representation through occlusion-aware voting, while an organ-part prior [15] confines the labels to complete leaf surfaces. To correct voting errors—where a single identity spans multiple leaves or one leaf is split by occlusion—we introduce a geometry-aware refinement. Small stray fragments are reassigned to the leaf whose surface they best continue, while over-segmented pieces are merged when their union forms a complete leaf and a long shared contact boundary (cut seam) indicates a common split.

We evaluate our framework on plant specimens with overlapping, touching, and folded leaves. Without any 3D supervision, it consistently recovers coherent per-leaf instances, reuniting folded leaves over-segmented by SAM 3 and resolving cross-view identity leakage between touching neighbors.

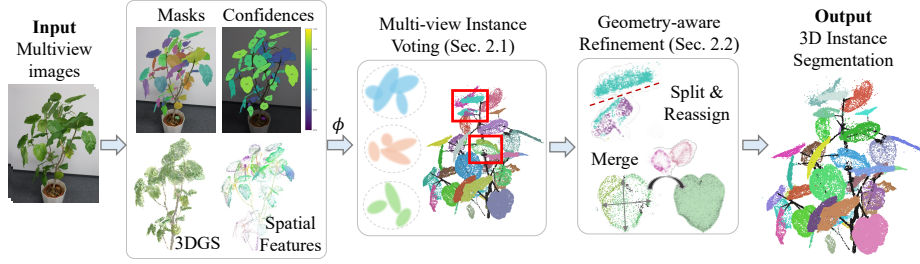


Fig. 1: Method pipeline. Given multi-view images of a plant, we first obtain a 3DGS reconstruction, spatial features [15], and SAM 3 [1] masks with confidences. These cues drive an occlusion-aware voting scheme that assigns initial per-leaf instances (Sec. 2.1), which a geometry-aware refinement then corrects using leaf shape completeness (Sec. 2.2), yielding the final 3D instance segmentation of the whole plant.

2 Methodology

2.1 Semantic-Guided Multi-View Plant Instance Voting

We obtain initial instances by prompting SAM 3 [1] with “leaf” in each view independently, so one leaf receives conflicting IDs across images. Inspired by geometry-driven multi-view voting [7, 14], we lift these masks to 3D and consolidate them into clean per-leaf labels, adapting the observation weighting to the dense self-occlusion of plant canopies. For Gaussian g_i of opacity α_i in view v , the alpha-compositing weight $\omega_{iv} = \alpha_i T_{iv}$ uses the accumulated transmittance $T_{iv} = \prod_{j \prec i} (1 - \alpha_j)$ of occluders in front. A binary indicator $o_{iv} \in \{0, 1\}$ discards Gaussians behind the front surface, preventing identity leakage across overlapping leaves. With mask confidence s_{iv} , the observation weight is $\phi_{iv} = \omega_{iv} o_{iv} s_{iv}$.

Each mask (v, m) then claims a weighted Gaussian set, with $c_{(v,m)}^i = \phi_{iv}$ if g_i projects into it and 0 otherwise. Their cross-view affinity is the soft Jaccard overlap

$$J((v, m), (v', m')) = \frac{\sum_i \min(c_{(v,m)}^i, c_{(v',m')}^i)}{\sum_i \max(c_{(v,m)}^i, c_{(v',m')}^i)}, \quad (1)$$

near one when both claim the same Gaussians. Agglomerative clustering on descending J —barring two masks from one view and discarding groups spanning too few views—groups them into global identities, each Gaussian adopting the identity y_i of greatest accumulated weight.

These votes are reliable in leaf interiors but noisy at boundaries. We thus leverage a coarse organ prior [15] to restrict labeling to the foliar surface $\mathcal{L}$; unvoted leaf Gaussians near the stem revert to stem, and the ambiguous band stays unlabeled. Within $\mathcal{L}$, we score each Gaussian by the visible contribution falling inside a mask, $r_i = \sum_v \phi_{iv} / \sum_v \omega_{iv} o_{iv}$ —high for consistently masked interiors, low at sparse margins. An Otsu threshold [10] on r_i isolates a confident core, and remaining leaf Gaussians inherit their nearest core label, giving the initial labeling $\{y_i\}$.

2.2 Geometry-Aware Refinement via Shape Completeness

Voting can inherit under-segmentation (one identity across several leaves) and over-segmentation (a fold splitting one leaf). We correct both in 3D: first redistributing stray fragments, then reuniting split leaves.

Fragment Reassignment. Leakage across a gap or occlusion hole can leave an instance in disconnected pieces. We thus split each instance—the Gaussians of identity k —into connected fragments by linking points within a small multiple of the median spacing $\bar{\rho}$. The largest fragment keeps the label; each residual fragment f is reassigned to the candidate leaf k' that best continues its surface over $\mathcal{N}(f) = \{k : d(f, k) \leq \delta\bar{\rho}\}$:

$$k' = \operatorname{argmin}_{k \in \mathcal{N}(f)} \frac{d(f, k)}{\bar{\rho}} + \beta \angle(\bar{n}_f, \bar{n}_k) + \gamma [\angle(\bar{n}_f, \bar{n}_k) < \theta_0] \|\bar{\psi}_f - \bar{\psi}_k\|, \quad (2)$$

with nearest-approach distance $d(f, k)$, mean normal $\bar{n}$, mean point embedding [15] $\bar{\psi}$, weights β, γ , and thresholds δ, θ_0 ; the bracket $[\cdot]$ activates the feature term only for coplanar pairs, which normals alone cannot separate. This favors a candidate that is near, surface-aligned, and feature-consistent. An empty $\mathcal{N}(f)$ spawns a new instance when f is leaf-sized, separating a leaked identity.

Shape-Complete Merging. To reunite a fold-split leaf, where local cues fail, we test whether a merged region forms one whole leaf. A point set P is *complete* when its shape matches the plant’s own leaves:

$$\text{Complete}(P) \triangleq [a(P) \leq a_{\max}] \wedge [\sigma(P) \geq \sigma_{\min}] \wedge [|P| \leq \eta], \quad (3)$$

where we flatten each leaf’s points onto a plane and measure its aspect ratio $a(P)$, solidity $\sigma(P)$ (occupied area over convex-hull area), and point count $|P|$; the size bound η is a multiple of the 90th-percentile segmented leaf size. Only a compact, filled, single-leaf region passes: a fold half fails alone but passes once joined to its complement. Since two touching leaves can also form a complete union, we merge adjacent instances only when their union is complete *and* their shared seam is long relative to leaf diameter, recovering folds without over-merging neighbors.

3 Experimental Results

We evaluate our method on the GaussianPlant dataset [13], using the three specimens with reliable instance ground truth, which differ in leaf density and overlap. We compare against HDBSCAN [9] applied directly to the 3DGS point cloud as a geometry-based unsupervised baseline, which separates instances by spatial density alone. We evaluate using Average Precision (AP), per-leaf recall at IoU 0.5 and 0.75, and the detected versus ground-truth leaf count, with instances matched one-to-one to the ground truth. As summarized in Tab. 1, our method consistently outperforms the baseline across all specimens, validating our overall pipeline design. This is further reflected in the qualitative comparison in

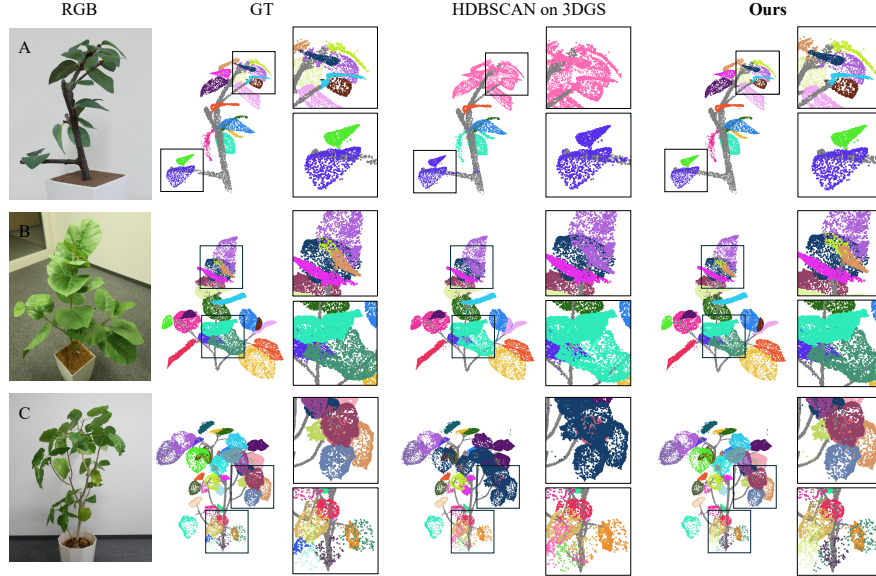


Fig. 2: Qualitative segmentation comparisons. From left to right: RGB image, ground truth, a naive 3DGS-based HDBSCAN baseline [9], and our result. Our method yields cleaner instance boundaries, as highlighted in the close-up insets.

Table 1: Quantitative comparison of leaf instance segmentation. We report AP, per-leaf recall at IoU 0.5/0.75, and the detected / ground-truth leaf count (Leaves).

Method	Plant A			Plant B			Plant C		
	AP $\uparrow$	R@0.5/0.75 $\uparrow$	Leaves	AP $\uparrow$	R@0.5/0.75 $\uparrow$	Leaves	AP $\uparrow$	R@0.5/0.75 $\uparrow$	Leaves
HDBSCAN [9]	0.189	0.278/0.167	6/18	0.667	0.762/0.667	16/21	0.454	0.562/0.458	34/48
Ours	0.883	1.000/0.944	18/18	0.914	0.952/0.952	21/21	0.919	0.958/0.938	46/48

Fig. 2: relying on spatial density alone, HDBSCAN fragments occluded leaves and merges touching ones, whereas our method accurately delineates individual leaves, recovering each as a coherent instance even where leaves touch.

4 Conclusion

We introduced an annotation-free framework for segmenting plants into individual leaves on 3D Gaussian reconstructions. Visibility-aware voting lifts per-view masks into consistent 3D identities, while geometry-aware refinement reassigns stray fragments and reunites over-segmented leaves without learned supervision. Yet, challenges remain—our current shape prior assumes compact leaf geometry, limiting its applicability to elongated foliage such as grasses. Extending the framework to accommodate a broader range of leaf morphologies is a promising direction for future work. Nevertheless, we believe our method provides a practical foundation for annotation-free 3D plant understanding and phenotyping.

Acknowledgment This work was in part supported by the Organization for the Promotion of Gender Equality at Nara Women’s University.

References

1. Carion, N., Gustafson, L., Hu, Y.T., Debnath, S., Hu, R., Suris, D., Ryali, C., Alwala, K.V., Khedr, H., Huang, A., et al.: Sam 3: Segment anything with concepts. arXiv preprint arXiv:2511.16719 (2025)
2. Fiorani, F., Schurr, U.: Future scenarios for plant phenotyping. *Annual Review of Plant Biology* **64**, 267–291 (2013)
3. Furbank, R.T., Tester, M.: Phenomics – technologies to relieve the phenotyping bottleneck. *Trends in Plant Science* **16**(12), 635–644 (2011)
4. Jiang, L., Zhao, H., Shi, S., Liu, S., Fu, C.W., Jia, J.: PointGroup: Dual-set point grouping for 3D instance segmentation. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2020)
5. Kerbl, B., Kopanas, G., Leimkühler, T., Drettakis, G., et al.: 3d gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.* **42**(4), 139–1 (2023)
6. Li, D., Li, J., Xiang, S., Pan, A.: Psegnet: Simultaneous semantic and instance segmentation for point clouds of plants. *Plant Phenomics* (2022)
7. Li, H., Wu, Y., Meng, J., Gao, Q., Zhang, Z., Wang, R., Zhang, J.: Instance-gaussian: Appearance-semantic joint gaussian representation for 3d instance-level perception. In: *Proceedings of the Computer Vision and Pattern Recognition Conference*. pp. 14078–14088 (2025)
8. Luo, L., Jiang, X., Yang, Y., Samy, E.R.A., Lefsrud, M., Hoyos-Villegas, V., Sun, S.: Eff-3DPSeg: 3D organ-level plant shoot segmentation using annotation-efficient deep learning. *Plant Phenomics* **5**, 0080 (2023). <https://doi.org/10.34133/plantphenomics.0080>
9. McInnes, L., Healy, J., Astels, S., et al.: hdbscan: Hierarchical density based clustering. *J. Open Source Softw.* **2**(11), 205 (2017)
10. Otsu, N.: A threshold selection method from gray-level histograms. *Automatica* **11**, 285–296 (1975)
11. Schult, J., Engelmann, F., Hermans, A., Litany, O., Tang, S., Leibe, B.: Mask3D: Mask transformer for 3D semantic instance segmentation. In: *IEEE International Conference on Robotics and Automation (ICRA)*. pp. 8216–8223 (2023)
12. Wang, D., Lu, X., Rinaldo, A.: DbSCAN: Optimal rates for density-based cluster estimation. *Journal of machine learning research* **20**(170), 1–50 (2019)
13. Yang, Y., Shinoda, R., Santo, H., Okura, F.: Gaussianplant: Structure-aligned gaussian splatting for 3d reconstruction of plants. arXiv preprint arXiv:2512.14087 (2025)
14. Ye, M., Danelljan, M., Yu, F., Ke, L.: Gaussian grouping: Segment and edit anything in 3d scenes. In: *European conference on computer vision*. pp. 162–179. Springer (2024)
15. Zhang, Y., Wu, X., Yang, Y., Fan, X., Li, H., Zhang, Y., Huang, Z., Wang, N., Zhao, H.: Utonia: Toward one encoder for all point clouds. arXiv preprint arXiv:2603.03283 (2026)